Lesson 23: Convex Sets

AI/ML Mathematics

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1 Motivation

Convex sets form the geometric foundation of optimization theory. The distinction between convex and non-convex feasible regions determines whether an optimization problem is tractable (solvable in polynomial time with global guarantees) or fundamentally hard (NP-hard in general). Every algorithm we study in later lessons—gradient descent, interior-point methods, proximal operators—relies on structural properties of convex sets established here.

RL/ML Connection

Convex sets appear throughout reinforcement learning and machine learning:

- **LP formulation of MDPs.** The set of valid occupancy measures for a Markov decision process is a polyhedron (intersection of finitely many halfspaces), and solving the MDP reduces to a linear program over this convex set.
- **Trust regions.** In TRPO and PPO the policy update is constrained to a KL-divergence ball around the current policy. This trust region is a convex set in the space of distributions.
- **Constraint sets.** Safety constraints in constrained MDPs (e.g., cost budgets) define convex feasible regions whose geometry governs dual decomposition methods.
- **Positive semidefinite cone.** Covariance matrices, kernel matrices, and semidefinite relaxations all live in the PSD cone, a fundamental convex cone.

Prerequisites. Familiarity with real analysis (limits, continuity, compactness) and linear algebra (inner products, eigenvalues, positive semidefiniteness) from Phases 00–01.

References. Boyd and Vandenberghe Boyd and Vandenberghe [2004], Chapter 2; Rockafellar Rockafellar [1970], Chapters 1–11.

2 Affine and Convex Sets

2.1 Lines and Affine Sets

Definition 2.1 (Affine Combination)**.** An *affine combination* of points $x_1, \ldots, x_k \in \mathbb{R}^n$ is any point of the form $\sum_{i=1}^k \theta_i x_i$ where $\sum_{i=1}^k \theta_i = 1$ (the weights θ_i may be negative).

Definition 2.2 (Affine Set)**.** A set $C \subseteq \mathbb{R}^n$ is *affine* if for every $x_1, x_2 \in C$ and every $\theta \in \mathbb{R}$, the point $\theta x_1 + (1-\theta)x_2 \in C$. Equivalently, C contains the entire line through any two of its points.

Remark 2.3. Every affine set can be written as $C = \{x_0 + v : v \in V\}$ for some subspace $V \subseteq \mathbb{R}^n$ and a point $x_0 \in C$. The subspace V is uniquely determined and equals $\{x - y : x, y \in C\}$. The *affine dimension* of C is $\dim V$.

2.2 Convex Sets

Definition 2.4 (Convex Set)**.** A set $C \subseteq \mathbb{R}^n$ is *convex* if for every $x_1, x_2 \in C$ and every $\theta \in [0,1]$:

$$\theta x_1 + (1-\theta)x_2 \in C.$$

Geometrically, the line segment between any two points of C lies entirely in C.

Remark 2.5. Every affine set is convex (since $[0,1] \subset \mathbb{R}$), but the converse fails: for example, the closed unit ball is convex but not affine.

2.3 Convex Combinations and the Convex Hull

Definition 2.6 (Convex Combination and Convex Hull)**.** A *convex combination* of points $x_1, \ldots, x_k$ is any point of the form $\sum_{i=1}^k \theta_i x_i$ where $\theta_i \geq 0$ and $\sum_{i=1}^k \theta_i = 1$. The *convex hull* of a set S is

$$\operatorname{conv}(S) = \left\{ \sum_{i=1}^k \theta_i x_i \;\middle|\; x_i \in S,\ \theta_i \geq 0,\ \sum_{i=1}^k \theta_i = 1,\ k \in \mathbb{N} \right\}.$$

Proposition 2.7 (Convex Hull is the Smallest Convex Superset)**.** *For any $S \subseteq \mathbb{R}^n$, the set* $\operatorname{conv}(S)$ *is convex, and* $\operatorname{conv}(S) \subseteq C$ *for every convex set C containing S.*

Proof. **Step 1: $\operatorname{conv}(S)$ is convex.** Let $x = \sum_{i=1}^k \alpha_i x_i$ and $y = \sum_{j=1}^m \beta_j y_j$ be two elements of $\operatorname{conv}(S)$, with $\alpha_i, \beta_j \geq 0$, $\sum_i \alpha_i = 1$, $\sum_j \beta_j = 1$. For $\theta \in [0,1]$,

$$\theta x + (1-\theta) y = \sum_{i=1}^k \theta \alpha_i \, x_i + \sum_{j=1}^m (1-\theta) \beta_j \, y_j.$$

The coefficients $\theta\alpha_i$ and $(1-\theta)\beta_j$ are nonnegative and sum to $\theta \cdot 1 + (1-\theta) \cdot 1 = 1$, so this is a convex combination of points in S. Hence $\theta x + (1-\theta) y \in \operatorname{conv}(S)$.

Step 2: $\operatorname{conv}(S)$ is contained in every convex superset. Let C be any convex set with $S \subseteq C$. We prove by induction on k that every convex combination of k points in S lies in C.

Base case ($k = 1$): $x_1 \in S \subseteq C$.

Inductive step: Given $\sum_{i=1}^{k+1} \theta_i x_i$ with $\theta_i \geq 0$, $\sum_i \theta_i = 1$. If $\theta_{k+1} = 1$ the point equals $x_{k+1} \in C$. Otherwise $\theta_{k+1} < 1$ and

$$\sum_{i=1}^{k+1} \theta_i x_i = (1-\theta_{k+1}) \sum_{i=1}^k \frac{\theta_i}{1-\theta_{k+1}} \, x_i \;+\; \theta_{k+1} \, x_{k+1}.$$

The inner sum is a convex combination of k points in S, hence in C by the inductive hypothesis. The outer expression is a convex combination of two points in C, hence in C by convexity. Therefore $\operatorname{conv}(S) \subseteq C$. □

2.4 Cones

Definition 2.8 (Cone and Convex Cone)**.** A set $C \subseteq \mathbb{R}^n$ is a *cone* if for every $x \in C$ and every $\theta \geq 0$, we have $\theta x \in C$. A set that is both convex and a cone is called a *convex cone*. Equivalently, C is a convex cone if and only if for all $x_1, x_2 \in C$ and all $\theta_1, \theta_2 \geq 0$:

$$\theta_1 x_1 + \theta_2 x_2 \in C.$$

Proof of the equivalence. ($\Rightarrow$) Let C be a convex cone, $x_1, x_2 \in C$, and $\theta_1, \theta_2 \geq 0$. If $\theta_1 + \theta_2 = 0$ the result is trivial ($0 \in C$ since $0 = 0 \cdot x_1$). Otherwise set $\lambda = \theta_1 + \theta_2 > 0$. Then $\theta_1 x_1 + \theta_2 x_2 = \lambda\big(\frac{\theta_1}{\lambda} x_1 + \frac{\theta_2}{\lambda} x_2\big)$. The inner term is a convex combination of $x_1, x_2 \in C$, hence in C. Scaling by $\lambda \geq 0$ preserves membership since C is a cone.

($\Leftarrow$) The cone property follows by setting $\theta_2 = 0$. Convexity follows because for $\theta \in [0,1]$, $\theta x_1 + (1-\theta) x_2$ has nonneg coefficients summing to 1. □

3 Important Examples

Example 3.1 (Hyperplanes and Halfspaces)**.** A *hyperplane* is a set of the form $\{x \in \mathbb{R}^n : a^T x = b\}$ for some $a \neq 0$ and $b \in \mathbb{R}$. Every hyperplane is affine (hence convex). A *halfspace* is $\{x \in \mathbb{R}^n : a^T x \leq b\}$, which is convex but not affine (unless $a = 0$).

Example 3.2 (Euclidean Balls and Ellipsoids)**.** The (closed) *Euclidean ball* centered at x_c with radius $r > 0$ is

$$B(x_c, r) = \{x \in \mathbb{R}^n : \|x - x_c\|_2 \leq r\}.$$

It is convex: for $x, y \in B(x_c, r)$ and $\theta \in [0, 1]$,

$$\|\theta x + (1-\theta)y - x_c\|_2 = \|\theta(x - x_c) + (1-\theta)(y - x_c)\|_2 \leq \theta\|x - x_c\|_2 + (1-\theta)\|y - x_c\|_2 \leq r.$$

More generally, an *ellipsoid* is $\mathcal{E} = \{x : (x - x_c)^T P^{-1}(x - x_c) \leq 1\}$ where $P \succ 0$. This equals $\{x_c + Pu : \|u\|_2 \leq 1\}$ (the affine image of the unit ball under $u \mapsto x_c + P^{1/2}u$), hence convex.

Example 3.3 (Polyhedra)**.** A *polyhedron* is the intersection of finitely many halfspaces and hyperplanes:

$$\mathcal{P} = \{x \in \mathbb{R}^n : Ax \leq b,\ Cx = d\},$$

where inequalities are componentwise. Since each halfspace and hyperplane is convex, and the intersection of convex sets is convex (Proposition 4.1 below), every polyhedron is convex. Bounded polyhedra are called *polytopes*.

Example 3.4 (The Positive Semidefinite Cone)**.** Let $\mathbb{S}^n$ denote the set of real $n \times n$ symmetric matrices. The *positive semidefinite (PSD) cone* is

$$\mathbb{S}^n_+ = \{X \in \mathbb{S}^n : X \succeq 0\} = \{X \in \mathbb{S}^n : v^T X v \geq 0 \text{ for all } v \in \mathbb{R}^n\}.$$

This is a convex cone. Indeed, if $A, B \succeq 0$ and $\theta_1, \theta_2 \geq 0$, then for any $v \in \mathbb{R}^n$:

$$v^T(\theta_1 A + \theta_2 B)v = \theta_1\, v^T A v + \theta_2\, v^T B v \geq 0.$$

Alternatively, $\mathbb{S}^n_+ = \bigcap_{v \in \mathbb{R}^n} \{X \in \mathbb{S}^n : v^T X v \geq 0\}$, an intersection of halfspaces in the space $\mathbb{S}^n \cong \mathbb{R}^{n(n+1)/2}$.

RL/ML Connection

The PSD cone $\mathbb{S}^n_+$ appears in covariance estimation (covariance matrices are PSD), in semidefinite programming relaxations for combinatorial problems, and in robust control for multi-agent systems where Lyapunov stability certificates require $P \succ 0$.

4 Operations Preserving Convexity

4.1 Intersection

Proposition 4.1 (Intersection Preserves Convexity)**.** *If $\{C_\alpha\}_{\alpha \in A}$ is any (possibly infinite) family of convex sets, then $\bigcap_{\alpha \in A} C_\alpha$ is convex.*

Proof. Let $x, y \in \bigcap_\alpha C_\alpha$ and $\theta \in [0, 1]$. For every α, $x, y \in C_\alpha$, so $\theta x + (1-\theta)y \in C_\alpha$ by convexity of C_α. Since this holds for all α, $\theta x + (1-\theta)y \in \bigcap_\alpha C_\alpha$. □

Remark 4.2. The intersection rule is extremely useful: it allows us to show convexity of complicated sets by writing them as intersections of simple convex sets. For instance, the PSD cone was expressed as an intersection of halfspaces in Example 3.4.

Warning

Unions of convex sets are **not** convex in general. For example, $\{0\} \cup \{1\} \subset \mathbb{R}$ is not convex.

4.2 Affine Image and Preimage

Proposition 4.3 (Affine Image and Preimage Preserve Convexity). *Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be an affine map, $f(x) = Ax + b$.*

(a) If $C \subseteq \mathbb{R}^n$ is convex, then $f(C) = \{Ax + b : x \in C\}$ is convex.

(b) If $D \subseteq \mathbb{R}^m$ is convex, then $f^{-1}(D) = \{x : Ax + b \in D\}$ is convex.

Proof. **(a) Image.** Let $u = Ax_1 + b$ and $v = Ax_2 + b$ be points in $f(C)$ with $x_1, x_2 \in C$. For $\theta \in [0, 1]$:

$$\theta u + (1 - \theta)v = A\big(\theta x_1 + (1 - \theta)x_2\big) + b = f\big(\theta x_1 + (1 - \theta)x_2\big).$$

Since $\theta x_1 + (1 - \theta)x_2 \in C$, we have $\theta u + (1 - \theta)v \in f(C)$.

(b) Preimage. Let $x_1, x_2 \in f^{-1}(D)$, so $Ax_1 + b$ and $Ax_2 + b$ lie in D. Then

$$A\big(\theta x_1 + (1 - \theta)x_2\big) + b = \theta(Ax_1 + b) + (1 - \theta)(Ax_2 + b) \in D,$$

by convexity of D. Hence $\theta x_1 + (1 - \theta)x_2 \in f^{-1}(D)$. □

Corollary 4.4. *The following operations preserve convexity:*

(i) ***Scaling and translation****: $\alpha C + a = \{\alpha x + a : x \in C\}$ for any $\alpha \in \mathbb{R}$, $a \in \mathbb{R}^n$.*

(ii) ***Projection****: If $C \subseteq \mathbb{R}^m \times \mathbb{R}^n$ is convex, then $\{x \in \mathbb{R}^m : (x, y) \in C$ for some $y \in \mathbb{R}^n\}$ is convex (as the image under $(x, y) \mapsto x$).*

(iii) ***Minkowski sum****: $C_1 + C_2 = \{x + y : x \in C_1,\ y \in C_2\}$ is convex whenever C_1, C_2 are convex (as the image of $C_1 \times C_2$ under $(x, y) \mapsto x + y$).*

4.3 Perspective Function

Proposition 4.5 (Perspective Function Preserves Convexity). *The* perspective function *$P : \mathbb{R}^{n+1} \to \mathbb{R}^n$ defined by $P(x, t) = x/t$ on $\operatorname{dom} P = \{(x, t) : t > 0\}$ preserves convexity in both directions:*

(a) If $C \subseteq \operatorname{dom} P$ is convex, then $P(C) \subseteq \mathbb{R}^n$ is convex.

(b) If $D \subseteq \mathbb{R}^n$ is convex, then $P^{-1}(D) = \{(x, t) : t > 0,\ x/t \in D\}$ is convex.

Proof. **(a)** Let $(x_1, t_1), (x_2, t_2) \in C$ with $t_1, t_2 > 0$ and $\theta \in [0, 1]$. Define $\bar{t} = \theta t_1 + (1 - \theta)t_2 > 0$. By convexity of C:

$$(\bar{x}, \bar{t}) := \big(\theta x_1 + (1 - \theta)x_2,\ \bar{t}\big) \in C.$$

Now compute

$$P(\bar{x}, \bar{t}) = \frac{\theta x_1 + (1 - \theta)x_2}{\bar{t}} = \frac{\theta t_1}{\bar{t}} \cdot \frac{x_1}{t_1} + \frac{(1 - \theta)t_2}{\bar{t}} \cdot \frac{x_2}{t_2}.$$

The coefficients $\theta t_1/\bar{t}$ and $(1 - \theta)t_2/\bar{t}$ are nonnegative and sum to 1, so this is a convex combination of $P(x_1, t_1)$ and $P(x_2, t_2)$. Hence $P(\bar{x}, \bar{t}) \in \operatorname{conv}(P(C)) = P(C)$ (the last equality holds because we have shown every convex combination of image points is an image point).

(b) Let $(x_1, t_1), (x_2, t_2) \in P^{-1}(D)$, so $x_1/t_1, x_2/t_2 \in D$ and $t_1, t_2 > 0$. The same algebraic identity shows $P(\theta(x_1, t_1) + (1 - \theta)(x_2, t_2))$ is a convex combination of x_1/t_1 and x_2/t_2, hence lies in D. Also $\bar{t} > 0$, so $(\bar{x}, \bar{t}) \in P^{-1}(D)$. □

4.4 Linear-Fractional Functions

Definition 4.6 (Linear-Fractional Function)**.** A *linear-fractional function* is $f : \mathbb{R}^n \to \mathbb{R}^m$ of the form

$$f(x) = \frac{Ax + b}{c^T x + d},$$

where $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$, $d \in \mathbb{R}$, and $\operatorname{dom} f = \{x : c^T x + d > 0\}$.

Proposition 4.7 (Linear-Fractional Functions Preserve Convexity)**.** *If C is a convex set contained in $\operatorname{dom} f$, then $f(C)$ is convex. If D is convex, then $f^{-1}(D) = \{x \in \operatorname{dom} f : f(x) \in D\}$ is convex.*

Proof. A linear-fractional function is the composition of an affine map $g : \mathbb{R}^n \to \mathbb{R}^{m+1}$ defined by $g(x) = (Ax + b,\ c^T x + d)$ with the perspective function $P(z, t) = z/t$. Both operations preserve convexity of images and preimages (Propositions 4.3 and 4.5), so their composition does as well. □

5 Separation Theorems

The separation theorems are the central results of this lesson. They provide the geometric foundation for Lagrangian duality: every dual variable corresponds to a separating hyperplane.

5.1 Separating Hyperplane Theorem

Theorem 5.1 (Separating Hyperplane Theorem)**.** *Let $C, D \subseteq \mathbb{R}^n$ be nonempty, disjoint, convex sets. Then there exist $a \in \mathbb{R}^n$ with $a \neq 0$ and $b \in \mathbb{R}$ such that*

$$a^T x \le b \quad \text{for all } x \in C, \qquad a^T x \ge b \quad \text{for all } x \in D.$$

Proof. We give a complete proof under the assumption that C is compact and D is closed (the fully general case requires the Hahn–Banach theorem from functional analysis).

Step 1: Existence of closest points. Define $\phi : C\times D \to \mathbb{R}$ by $\phi(x, y) = \|x-y\|_2$. Since C is compact and D is closed, and $\phi(x, y) \to \infty$ as $\|y\| \to \infty$, the infimum $d^* = \inf_{x\in C, y\in D} \|x-y\|_2$ is attained at some pair (x^*, y^*). Since $C \cap D = \emptyset$, $d^* > 0$.

Step 2: Construction of the separating hyperplane. Set $a = y^* - x^*$ and $b = \frac{1}{2}(a^T x^* + a^T y^*)$. We claim: $a^T x \le a^T x^*$ for all $x \in C$, and $a^T y \ge a^T y^*$ for all $y \in D$. Since $a^T x^* < a^T y^*$ (as $a^T(y^* - x^*) = \|a\|_2^2 > 0$), the value b separates C and D.

Step 3: Proof for C. Let $x \in C$ and define $g(t) = \|x^* + t(x - x^*) - y^*\|_2^2$ for $t \in [0, 1]$. Since C is convex, $x^* + t(x - x^*) \in C$ for all $t \in [0, 1]$. By the minimality of (x^*, y^*):

$$g(t) = \|x^* + t(x - x^*) - y^*\|_2^2 \ge \|x^* - y^*\|_2^2 = g(0) \quad \text{for all } t \in [0, 1].$$

The function g is differentiable, so $g'(0) \ge 0$. Computing:

$$g'(0) = 2(x - x^*)^T(x^* - y^*) = -2(x - x^*)^T a \ge 0.$$

Rearranging: $a^T x \le a^T x^*$.

Step 4: Proof for D. Let $y \in D$ and define $h(t) = \|x^* - y^* - t(y - y^*)\|_2^2$ for $t \in [0, 1]$. Since D is convex, $y^* + t(y - y^*) \in D$, and by minimality:

$$h(t) = \|x^* - (y^* + t(y - y^*))\|_2^2 \ge \|x^* - y^*\|_2^2 = h(0).$$

So $h'(0) \ge 0$:

$$h'(0) = -2(y - y^*)^T(x^* - y^*) = 2(y - y^*)^T a \ge 0.$$

Rearranging: $a^T y \ge a^T y^*$.

Conclusion. We have $a^T x \le a^T x^*$ for all $x \in C$ and $a^T y \ge a^T y^*$ for all $y \in D$, with $a^T x^* < b < a^T y^*$. In particular, $a^T x \le b$ for all $x \in C$ and $a^T y \ge b$ for all $y \in D$. □

Key Result

The separating hyperplane theorem is the geometric engine behind **Lagrangian duality**. Every Lagrange multiplier defines a separating hyperplane between the feasible set (in objective-constraint space) and the half-plane of strictly better objectives. Strong duality holds precisely when this separation is "tight."

5.2 Supporting Hyperplane Theorem

Theorem 5.2 (Supporting Hyperplane Theorem). *Let $C \subseteq \mathbb{R}^n$ be a nonempty convex set and let $x_0 \in \partial C$ (the boundary of C). Then there exists $a \in \mathbb{R}^n$ with $a \neq 0$ such that*

$$a^T x \leq a^T x_0 \quad \textit{for all } x \in C.$$

The hyperplane $\{x : a^T x = a^T x_0\}$ is called a supporting hyperplane *to C at x_0.*

Proof. Since $x_0 \in \partial C$, there exists a sequence $\{y_k\}_{k=1}^{\infty}$ with $y_k \notin C$ and $y_k \to x_0$ as $k \to \infty$.

For each k, the singleton $\{y_k\}$ and the convex set C are disjoint. Apply the separating hyperplane theorem (with $\{y_k\}$ compact and C closed—if C is not closed, apply to $\mathrm{cl}(C)$ which is also convex and has the same boundary): there exists $a_k \in \mathbb{R}^n$ with $\|a_k\|_2 = 1$ (normalize) such that

$$a_k^T x \leq a_k^T y_k \quad \text{for all } x \in C.$$

The sequence $\{a_k\}$ lies on the unit sphere S^{n-1}, which is compact. By the Bolzano–Weierstrass theorem, there is a convergent subsequence $a_{k_j} \to a^*$ with $\|a^*\|_2 = 1$.

Passing to the limit in $a_{k_j}^T x \leq a_{k_j}^T y_{k_j}$:

$$(a^*)^T x \leq (a^*)^T x_0 \quad \text{for all } x \in C.$$

Here we used $a_{k_j} \to a^*$ and $y_{k_j} \to x_0$ (the subsequence of $\{y_k\}$ also converges to x_0). Setting $a = a^*$ completes the proof. □

RL/ML Connection

The supporting hyperplane theorem guarantees that at every boundary point of a convex feasible set, there is a "tangent" linear constraint. In constrained policy optimization (CPO), the safety-constraint boundary is approximated by such a supporting hyperplane, yielding a linear constraint on the policy update that can be solved efficiently.

6 Exercises

Exercise 6.1 (Convexity of Set Operations). Let $C_1, C_2 \subseteq \mathbb{R}^n$ be convex. Prove that each of the following is convex:

(a) The Minkowski sum $C_1 + C_2 = \{x + y : x \in C_1,\ y \in C_2\}$.

(b) The set $\{(\lambda, x) \in \mathbb{R} \times \mathbb{R}^n : x \in \lambda C_1,\ \lambda \geq 0\}$.

(c) The set difference $C_1 \setminus C_2$ is **not** convex in general. Give a counterexample in $\mathbb{R}^2$.

Exercise 6.2 (Separating Hyperplane for Disk and Point). Let $C = \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1\}$ and $p = (2, 0)$.

(a) Find the closest point $x^* \in C$ to p and use the construction from Theorem 5.1 to write down the separating hyperplane explicitly.

(b) Verify that the separating hyperplane from part (a) is also a supporting hyperplane to C at x^*.

Exercise 6.3 (Convex Hull of Finitely Many Points)**.** Let $S = \{e_1, \ldots, e_n\} \subset \mathbb{R}^n$ be the standard basis vectors. Show that $\operatorname{conv}(S) = \Delta_n$, the probability simplex $\{x \in \mathbb{R}^n : x_i \geq 0,\ \sum_i x_i = 1\}$.

Exercise 6.4 (PSD Cone Properties)**.** (a) Show that $\mathbb{S}^n_+$ is a closed set (in the topology of $\mathbb{S}^n \cong \mathbb{R}^{n(n+1)/2}$). *Hint:* Write $\mathbb{S}^n_+$ as an intersection of closed sets.

(b) Show that $\mathbb{S}^n_+$ is a *proper* cone: it is convex, closed, solid (nonempty interior), and pointed ($X \in \mathbb{S}^n_+$ and $-X \in \mathbb{S}^n_+$ implies $X = 0$).

(c) Prove that the interior of $\mathbb{S}^n_+$ is $\mathbb{S}^n_{++} = \{X \in \mathbb{S}^n : X \succ 0\}$.

Exercise 6.5 (Perspective Function Application)**.** Let $D = \{(x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1\}$ (the unit disk). Compute the set $P^{-1}(D) = \{(x_1, x_2, t) \in \mathbb{R}^3 : t > 0,\ (x_1/t)^2 + (x_2/t)^2 \leq 1\}$ and show directly (without using Proposition 4.5) that it is convex.

References

Stephen Boyd and Lieven Vandenberghe. *Convex Optimization.* Cambridge University Press, 2004.

R. Tyrrell Rockafellar. *Convex Analysis.* Princeton University Press, 1970.